



Whole Genome De novo Sequencing Report

2024.4

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Order Information

Client Name	MacroGen
Client Organization	MacroGen HQ
Order Number	HN000000000, HN000000001
Application	Whole Genome De novo Sequencing
Library Type	[3.0] PacBio Microbial Library, TruSeq Nano DNA (350)

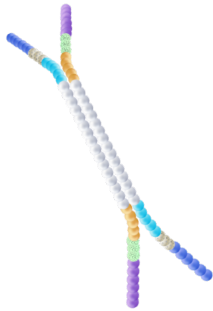
Illumina Experimental Workflow

The samples are prepared according to NGS library preparation workflow, and sequenced using Illumina platform. The workflow illustrated below shows the common ligation based method of library preparation. The process may differ based on the library preparation protocol followed.



Sample Preparation

DNA/RNA is first extracted from the sample, and samples which meet quality control standards proceed to library construction.



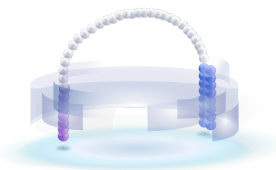
Ligate Adapters

The sequencing library is prepared by random fragmentation of the DNA or cDNA sample, followed by 5' and 3' adapter ligation. Alternatively, "tagmentation" combines the fragmentation and ligation reactions into a single step which greatly increases the efficiency of the library preparation process.

Final library Construction

Adapter-ligated fragments are then PCR amplified with a PCR primer solution which anneals to the ends of each adapters.

The library templates undergo quality control and quantification process.

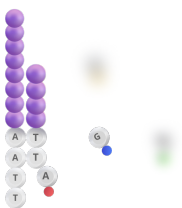


Cluster generation using bridge amplification

The library is loaded onto a flow cell where fragments are captured on a lawn of surface-bound oligos complementary to the library adapters.

Each fragment is then amplified into distinct clonal clusters through bridge amplification.

Once cluster generation is complete, the templates are ready for sequencing.



Sequencing by synthesis (SBS) technology

Illumina SBS technology utilizes a proprietary reversible terminator-based method that detects single bases as they are incorporated into DNA template strands. As all 4-reversible, terminator-bound dNTPs are present during each sequencing cycle, natural competition minimizes incorporation bias and greatly reduces raw error rates compared to other technologies.

The result is highly accurate base-by-base sequencing that virtually eliminates sequence-context-specific errors, even within repetitive sequence regions and homopolymers.



Generation of Raw data

The Illumina sequencer generates raw images utilizing sequencing control software for system control and base calling, through integrated primary analysis software called RTA (Real Time Analysis).

The BCL/cBCL (base call) binary files are converted into FASTQ files using bcl2fastq, which is an Illumina provided package. Adapters are not trimmed away from the reads.

PacBio Experimental Workflow

The samples are prepared according to a guide for preparing SMRTbell template for sequencing on the PacBio instrument.



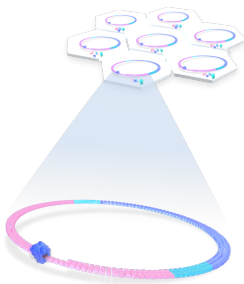
Sample Preparation

DNA/RNA is first extracted from the sample, and samples which meet quality control standards proceed to library construction.



Adapter Ligation

The SMRTbell library preparation creates a circularized template by ligating hairpin adapters to the end of a high-quality double stranded DNA (dsDNA) molecule. After ligation, sequencing primer and sequencing polymerase are annealed to the library, making it ready for SMRT Sequencing.



SMRT(Single Molecule, Real-Time) Sequencing

Single Molecule, Real-Time (SMRT) Sequencing is the core technology powering PacBio long-read sequencing platforms.

When the SMRTbells are loaded to a SMRT cell, a SMRTbell diffuses into a zero-mode waveguide (ZMW, sequencing unit). In each ZMW, a single polymerase is immobilized at the bottom, which can bind to either hairpin adaptor of the SMRTbell and start the replication. Four fluorescent -labeled nucleotides, which generate distinct emission spectrums, are added to the SMRT cell. As a base is held by the polymerase, a light pulse is produced that identifies the base.

SMRT cell

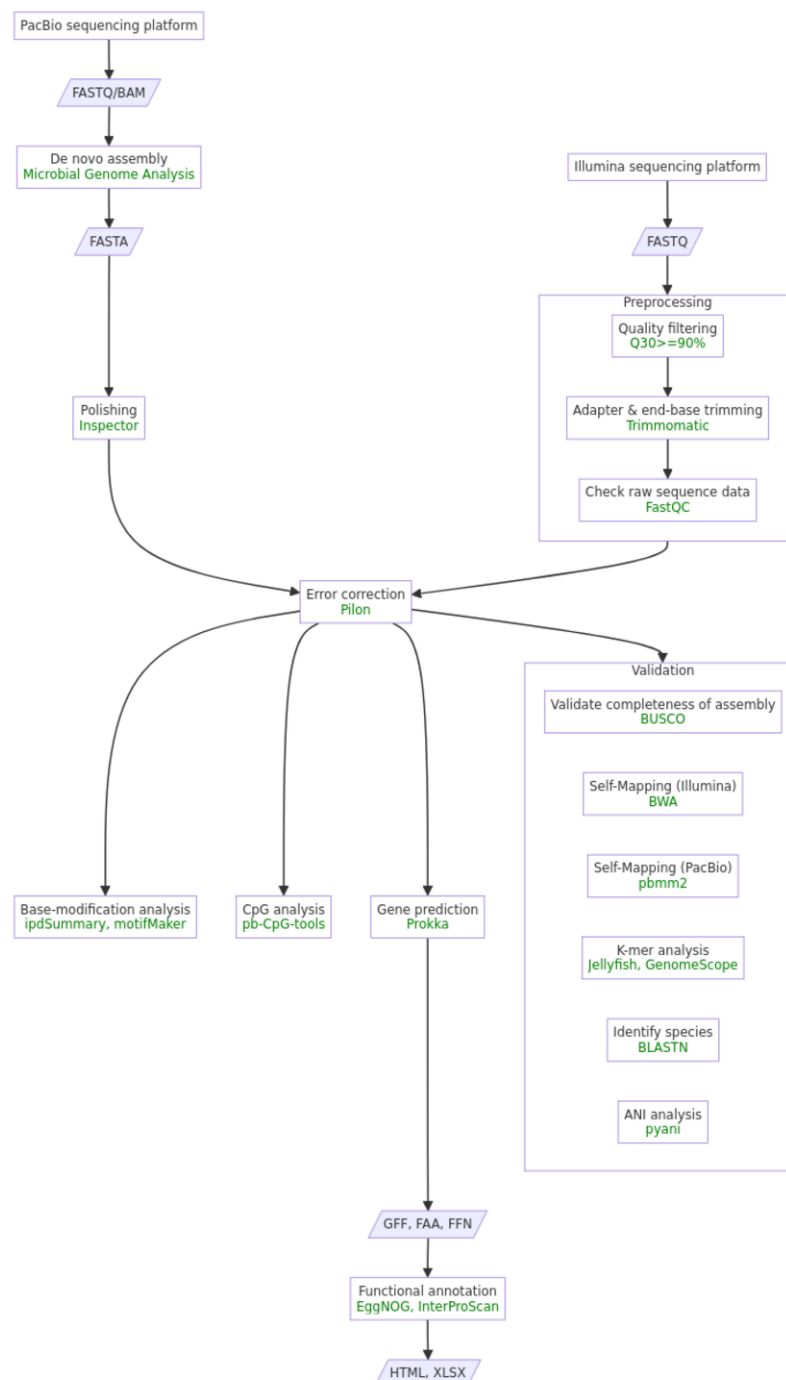
The replication processes in all ZMWs of a SMRT cell are recorded by a “movie” of light pulses, and the pulses corresponding to each ZMW can be interpreted to be a sequence of bases. Because the SMRTbell forms a closed circle, after the polymerase replicates one strand of the target dsDNA, it can continue incorporating bases of the adapter and then the other strand. If the lifetime of the polymerase is long enough, both strands can be sequenced multiple times (called “passes”).



Generation of Raw data

Performs real-time signal processing and base calling using Primary Analysis software installed on the instrument.

Analysis Workflow



Pre-processing

- Quality Control : After sequencing, the raw reads undergo quality control. Overall quality of reads generated, total bases, total reads, GC content and basic statistics are calculated. The data can be used to check raw data whether there are any problems in raw data or not before doing analysis.
- Preprocessing : When performing sequencing, if the read length is longer than an insert size, there may be sequences corresponding to the adapter in 3' end of sequencing reads. In order to reduce biases in analysis, adapter trimming and quality filtering are performed. The quality of filtered reads, total bases, total reads, GC content and basic statistics are calculated again.

Assembly

De novo assembly is performed. When stats of the assembly results is poor, other assembler is tried to get better results.

Error Correction

After de novo assembly step, high accuracy reads can apply for sequence compensation to construct contigs more accurately. By mapping high accuracy reads to first assembled genome sequence, we can see the mapping result that shows a slight difference from the assembly result. We use this information to correct the consensus sequence.

Base-modification Analysis

DNA base modifications are predicted using inter-pulse duration (IPD) information from the generated PacBio data.

CpG Analysis

The probability of methylation of each site is calculated by mapping raw data containing the 5mC base modification tags against the primary contigs/scaffolds.

Assembly Validation

The assembled genome is validated using K-mer analysis, self-mapping strategy, BLASTN, ANI analysis and BUSCO analysis.

K-mer analysis : This process provides information of k-mer coverage, heterozygosity and estimated genome size.

Self-mapping (Illumina) : Raw (or preprocessed) reads are aligned against assembled genome and their insert size are estimated for validation.

Self-mapping (PacBio) : In order to validate accuracy of the assembly, raw (or preprocessed) reads are aligned against assembled genome and calculate how many reads were used in assembly.

BLAST : The assembled genome is subjected to BLASTN against NCBI nucleotide database. By the results, we can figure out whether the assembled contigs is concordant with sample information provided by customer or not.

ANI analysis : For prokaryote, average nucleotide identity (ANI) is calculated for biological identification.

BUSCO analysis : BUSCO analysis is performed to evaluate genome assembly based on evolutionarily-informed expectations of gene contents.

Gene Prediction & Annotation

After whole genome or draft genome is assembled, the location of protein-coding sequences, tRNA genes, rRNA genes, and tmRNA genes are identified. Then their functions are annotated.

Raw Data Statistics

The total number of bases, reads, GC (%), Q20 (%), and Q30 (%) are calculated for the 1 sample(s).
For example, in SAMPLE1 sample, 25,000,000 reads are produced, and total read bases are 3.5 Gbp.
The GC content (%) is 70.0% and Q30 is 90.0%.

* Raw Data

Sample ID	Total bases (bp)	Total reads	GC (%)	AT (%)	Q20 (%)	Q30 (%)
SAMPLE1	3,500,000,000	25,000,000	70.0	30.0	95.0	90.0
SAMPLE1_filtered	2,000,000,000	15,000,000	70.0	30.0	99.0	95.0

- Sample ID : Sample name.
- Total Bases (bp) : Total number of bases sequenced.
- Total Reads : Total number of reads. For illumina paired-end sequencing, this value refers to the sum of read1 and read2.
- GC (%) : Ratio of GC content.
- AT (%) : Ratio of AT content.
- Q20 (%) : Ratio of bases that have phred quality score of over 20.
- Q30 (%) : Ratio of bases that have phred quality score of over 30.

HiFi Read Statistics

The total number of bases, HiFi reads, N50, average read length, average read quality, and average pass are calculated for the 1 sample(s). For example, in SAMPLE1 sample, 30,000 HiFi reads are produced, and total read bases are 250 Mbp. The N50 is 10,000 bp, average read length is 9,000bp, average quality is Q35, and average pass is 15.

Sample ID	HiFi Read Bases	HiFi Reads	HiFi N50	Average Read Length	Average Read Quality	Average Pass
SAMPLE1	250,000,000	30,000	10,000	9,000	Q35	15

- Sample ID : Sample name.
- HiFi Read Bases : Total bases of HiFi reads.
- HiFi Reads : The number of reads in the HiFi read.
- HiFi Read N50 : 50% of all bases come from HiFi reads longer than this value.
- Average Read Length : The mean length of HiFi reads.
- Average Read Quality : The mean quality of HiFi reads.
- Average Pass : The mean number of passes in the HiFi reads.

Summary of Project Results

Summary of de novo Assembly

Assembly Method	Contigs	Total Contig Bases	N50	GC Content (%)	Estimated Genome Size	Sequencing Depth
Microbial Genome Analysis	1	8,000,000	8,000,000	65.0	8,000,000	31.25

Contigs : The number of contigs assembled.

Total Contig Bases : The total length of contigs.

N50 : 50% of all bases come from contigs longer than this value.

GC Content (%) : GC content of assembly result.

Estimated Genome Size : Estimated genome size of the target derived from K-mer frequency analysis result.

Sequencing Depth : Total read bases per genome assembly result.

Taxon Identification Using ANI Analysis

Closest Genome	Average Nucleotide Identity	Alignment Coverage
GCF_000000000.1	1.00	0.50

Summary of Gene Prediction and Annotation

Sample	Contigs	Bases	Gene	CDS	tRNA	rRNA	tmRNA
SAMPLE1	1	8,000,000	5,700	5,500	80	20	1

Sample : The sample name.

Contigs : The number of contigs in assembled sequence.

Bases : The number of bases in assembled sequence.

Gene : The number of genes in assembled sequence.

CDS : The number of CDSs in assembled sequence.

tRNA : The number of tRNAs in assembled sequence.

rRNA : The number of rRNAs in assembled sequence.

tmRNA : The number of tmRNAs in assembled sequence.

De novo Assembly

Contig

Bioinformatics software such as Microbial Genome Analysis, CANU, Flye and Raven can assemble the PacBio long-reads. In this analysis, Microbial Genome Analysis was used and the detailed information is attached in the appendix. As the result of mapping reads against assembled contigs and error correction using Racon, the consensus sequence with higher quality is generated. Furthermore, Microbial Genome Analysis rotated contig start sites as origin of replication for each contig. The assembly results are summarized in the table below.

Summary of Assembly

Number of Contigs	Total Contig Bases	N50	Max Length	Min Length	Mean Length
1	7,999,990	7,999,990	7,999,990	7,999,990	7,999,990

Number of Contigs : The number of contigs assembled.

Total Contig Bases : The total length of contigs.

N50 : 50% of all bases come from contigs longer than this value.

Max Length : The length of maximum contig.

Min Length : The length of minimum contig.

Mean Length : The average length of contigs assembled.

After assembly, Illumina reads were applied for accurate genome sequence using Pilon. And then, by mapping the hifi reads against assembled contigs, the consensus sequence with depth of coverage data was generated.

Summary of Corrected Assembly

Number of Contigs	Total Contig Bases	N50	Max Length	Min Length	Mean Length
1	8,000,000	8,000,000	8,000,000	8,000,000	8,000,000

Number of Contigs : The number of contigs assembled.

Total Contig Bases : The total length of contigs.

N50 : 50% of all bases come from contigs longer than this value.

Max Length : The length of maximum contig.

Min Length : The length of minimum contig.

Mean Length : The average length of contigs assembled.

Result of Assembly : single contig was formed

Contig Name	Length	GC (%)	Depth	Circular	Alias
contig1	8,000,000	65.0	35	YES	Contig1

Length : The number of bases in each contig

GC(%) : GC content

Depth : The average number of reads aligned to each of the bases on contig

Circular : 5' end and 3' end are connected

Alias : The alias is named based on the BLASTN resultThe following two conditions are used to create an alias:

a : Query cover 80 or more

b : Similarity between genome size

If both conditions are met, it is named Chromosome or Plasmid

If not, named it Contig

Contig Information

Nx Stat of Contigs

	Average Length		Cumulative Sum	Number of Contigs
N10	8,000,000	8,000,000	8,000,000	1
N20	8,000,000	8,000,000	8,000,000	1
N30	8,000,000	8,000,000	8,000,000	1
N40	8,000,000	8,000,000	8,000,000	1
N50	8,000,000	8,000,000	8,000,000	1
N60	8,000,000	8,000,000	8,000,000	1
N70	8,000,000	8,000,000	8,000,000	1
N80	8,000,000	8,000,000	8,000,000	1
N90	8,000,000	8,000,000	8,000,000	1
N100	8,000,000	8,000,000	8,000,000	1

Average Length : The average length in the contigs statistics (N10~N100).

Cumulative Sum : The sum of the length in the contigs statistics (N10~N100).

Number of Contigs : The number of sequences in the contigs statistics (N10~N100).

Base Contents of Contigs

Num of A	Num of T	Num of G	Num of C	Num of N	GC Contents (%)
900,000	900,000	2,600,000	2,500,000	0	65.0

Num of A : The total number of adenine (A) bases

Num of T : The total number of thymine (T) bases

Num of G : The total number of guanine (G) bases

Num of C : The total number of cytosine (C) bases

Num of N : The total number of ambiguous bases

GC Contents : The percentage of guanine-cytosine base pairs.

Assembly Validation

K-mer Analysis

K-mer analysis was performed to estimate the genome size of sample. The graph was plotted with the coverage and frequency of k-mers. The sharp left-side peak represents random sequencing error while the right represents appropriate data. The genome size can be estimated using total k-mer number and volume peak.

21-mer

K-mer Coverage	Heterozygosity	Genome Length	Genome Repeat Length
40.00	0.500	8,000,000	1,000

K-mer Coverage : The mean k-mer coverage for heterozygous bases.

Heterozygosity : The overall rate of heterozygosity.

Genome Length : The inferred genome length.

Genome Repeat Length : The length of the genome that is repetitive.

Self-Mapping Results

In order to identify how much reads were used in assembly, raw (or preprocessed) data reads were mapped to an assembly result. After mapping, the necessary statistics were calculated.

Overall Mapping Stats

Library Name	Total Reads	Mapped Reads	Coverage (%)	Depth	Insert Size (Std.)
SAMPLE1 (Illumina)	15,000,000	14,999,000 (99.99%)	99.99	300.00	450.00 (80.00)
SAMPLE1 (PacBio)	25,000	24,999 (99.99%)	100.00	35.00	-

Library Name : Sample's library name

Total Reads : Total number of reads

Mapped Reads : Total number of mapped reads.

Coverage (%) : The percentage of mapped sites ($\geq 1x$).

Depth : Average mapping depth.

Insert Size (Std.) : The length between adapters and standard deviation of predicted length.

BLAST Results

After complete genome or draft genome was assembled, BLAST analysis was carried out to identify to which species each contig show similarity. Best hit results were identified using NCBI NT database. Each result was prepared separately by the sheet of excel. Following figure shows the proportion based on the genus level as the result of the best hit for the entire contig.

BLAST Result of contig

Contig	Contig Length	Subject Description	Subject Length	E-value
contig1	8,000,000	CP000000.1 TEST SPECIES	8,000,000	0.0

Contig : The name of contig.

Contig Length : The total number of bases in the contig.

Subject Description : Description of sequence matched by BLASTn.

Subject Length (bp) : Length of sequence matched by BLASTn.

E-value : The expectations that could be matched by chance. The lower, the more significant it is.

Because the BLAST analysis is based on registered information, it is difficult to determine the information of the species that is not registered. In particular, the assembly results could be matched with a relative species or an evolutionarily distant species due to sequence differences or error that may occur during the assembly process. Therefore, it would be more appropriate to use the analysis results to identify patterns rather than to use it as an absolute criterion for species determination.

BUSCO Results

In order to assess the completeness of the genome assembly, Benchmarking Universal Single-Copy Orthologs (BUSCO) analysis was performed based on evolutionarily-informed expectations of gene content from near-universal single-copy orthologs. The recovered matches are classified as 'Complete' if their lengths are within the expectation of the BUSCO profile match lengths. If these are found more than once, they are classified as 'duplicated'. The matches that are only partially recovered are classified as 'Fragmented', and BUSCO groups for which there are no matches that pass the tests of orthology are classified as 'Missing'. Higher complete BUSCOs may indicate good assembly. However for species other than model organisms, relatively low BUSCOs can be appears due to characteristics of the sample as well as the incompleteness of the assembly. By default, bacteria or eukaryota DB was used for analysis.

Used Lineage : bacteria_odb10 (number of genomes: 4085, number of BUSCOs: 124)

BUSCO Analysis Result

Status	Number of BUSCOs	Percentage (%)
Complete BUSCOs (C)		
Complete and single-copy BUSCOs (S)	120	96.77
Complete and duplicated BUSCOs (D)	4	3.23
Fragmented BUSCOs (F)	0	0.00
Missing BUSCOs (M)	0	0.00
Total BUSCO groups searched	124	

Status : A quantitative assessment list of the completeness in terms of expected gene content. The following two conditions are used to create a status:

a : Expected range of scores

b : Expected range of length alignments

If both conditions are met, it is classified as Complete (These complete busco matches are either single-copy or duplicated).

If both conditions are not met, it is classified as Missing.

Number of BUSCOs : Identified count in sample

Percentage (%) : Identified percentage in sample.

ANI Analysis

Average Nucleotide Identity (ANI) is a measure of nucleotide level genomic similarity between the same taxon species available in the NCBI database. To identify species or/and strain information, ANI analysis is performed with taxon of the reference sequence matched with contig1 by BLASTN.

Used taxonomy ID : 1000000

Top 1 Genomes Resulted by ANI Analysis

Ranking	Similar Genome	ANI (%)	Aln. Cov. (%)
1	GCF_000000000.1	100.00	50.00

Ranking : The order of the highest ANI value.

Similar Genome : Species name which has high ANI value.

ANI (%) : % of relatedness at whole-genome level.

Aln. Cov. (%) : Alignment coverage. % of coverage by assembly sequence alignments against compared genome.

Gene Prediction and Annotation

Gene Contents

After complete genome or draft genome was assembled, the locations of protein genes were predicted and their functions were annotated. Prokka was used to predict the location while BLAST was used when want to know about function and identification of assembled sequences against nucleotide and protein sequences database.

Summary of Gene Prediction

Sample	Contigs	Bases	Gene	CDS	tRNA	rRNA	tmRNA
SAMPLE1	1	8,000,000	5,700	5,500	80	20	1

Sample : The sample name.

Contigs : The number of contigs in assembled sequence.

Bases : The number of bases in assembled sequence.

Gene : The number of genes in assembled sequence.

CDS : The number of CDSs in assembled sequence.

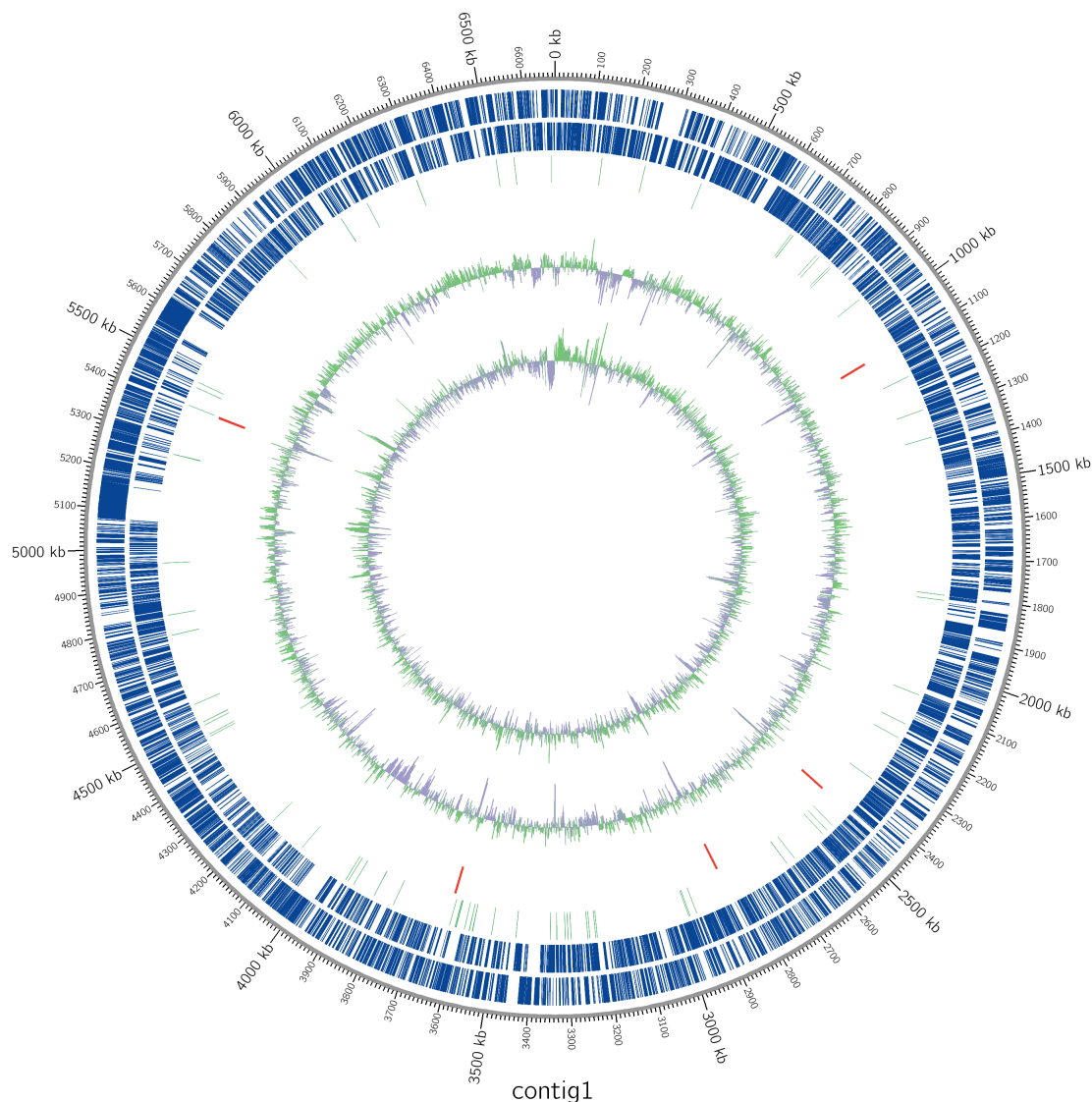
tRNA : The number of tRNAs in assembled sequence.

rRNA : The number of rRNAs in assembled sequence.

tmRNA : The number of tmRNAs in assembled sequence.

Circular Map

Circular map was generated based on the annotation of the contig(s). Marked characteristics are shown from outside to the center; CDS on forward strand, CDS on reverse strand, tRNA, rRNA, GC content and GC skew.



Forward CDS : Region of forward CDS, region that is not CDS is described as blank.

Reverse CDS : Region of reverse CDS, region that is not CDS is described as blank.

tRNA : Region of tRNA is marked in light green.

rRNA : Region of rRNA is marked in red.

GC content : Region that has higher value of GC percentage than average is described in exterior light green peak. Otherwise, it is described in interior lavender peak. Height of the peak describes the difference from the average GC percentage.

GC skew : According to the formula, $(G-C)/(G+C)$, positive value shows that G is dominant while negative value shows that C is dominant. The exterior light green peak describes the region that has higher G content while interior lavender peak describes the region that higher C content.

* : Please use this information only as reference since most of the time it is hard to discriminate origin and terminus from mapping.

Functional Annotation

EggNOG Annotation

EggNOG	Description	Count	Ratio (%)
A	RNA processing and modification	1	0.0185
B	Chromatin structure and dynamics	1	0.0185
C	Energy production and conversion	228	4.2183
D	Cell cycle control, cell division, chromosome partitioning	26	0.4810
E	Amino acid transport and metabolism	318	5.8834
F	Nucleotide transport and metabolism	71	1.3136
G	Carbohydrate transport and metabolism	363	6.7160
H	Coenzyme transport and metabolism	116	2.1462
I	Lipid transport and metabolism	121	2.2387
J	Translation, ribosomal structure and biogenesis	167	3.0897
K	Transcription	468	8.6586
L	Replication, recombination and repair	175	3.2377
M	Cell wall/membrane/envelope biogenesis	176	3.2562
N	Cell motility	1	0.0185
O	Posttranslational modification, protein turnover, chaperones	114	2.1092
P	Inorganic ion transport and metabolism	264	4.8844
Q	Secondary metabolites biosynthesis, transport and catabolism	144	2.6642
R	General function prediction only	669	12.3774
S	Function unknown	1599	29.5837
T	Signal transduction mechanisms	255	4.7179
U	Intracellular trafficking, secretion, and vesicular transport	32	0.5920
V	Defense mechanisms	96	1.7761
W	Extracellular structures	0	0.0000
Y	Nuclear structure	0	0.0000
Z	Cytoskeleton	0	0.0000

Base-modification Analysis

DNA Base-modification such as m4C and m6A could be predicted using inter-pulse duration (IPD) information of PacBio data generated from PacBio sequencer. ipdSummary, a bundle tool of SMRT Link, loaded IPDs observed at each position in the genome, compared those IPDs to value expected for unmodified DNA, and outputs the result of this statistical test. From the result of ipdSummary, motifMaker, also bundle tool of SMRT Link, annotated motifs associated with DNA modification.

* SAMPLE1

Statistics for the Methyltransferase Recognition Motifs Detected

Motif	Position	Type	# Detected	# of Motifs in Genome	Mean QV	Mean Coverage
BGNCCGCGGR	4	m4C	300	1,500	40.00	35.01
GGCCCSGA	5	m4C	2,000	2,005	65.00	30.00
VGNCTGVDCR	4	m4C	1,000	10,901	45.55	35.00

Motif : The nucleotide sequence of the methyltransferase recognition motif, using the standard IUPAC nucleotide alphabet

Position : The position within the motif that is modified. Example: The modified adenine in GATC is at position 2

Type : The type of chemical modification most commonly identified at that motif. These are: 6mA, 4mC, 5mC or modified_base (modification not recognized by the software.)

Detected : The number of times that this motif was detected as modified across the entire genome

of Motifs in Genome : The number of times this motif occurs in the genome

Mean QV : The mean modification QV for all instances where this motif was detected as modified

Mean Coverage : The mean coverage for all instances where this motif was detected as modified

CpG Analysis

CpG/5mC methylations were predicted using pb-CpG-Tools. Raw data which contain 5mC base modification tags generated by Jasmine were mapped against primary contigs and methylation probability of each site was calculated.

Result of CpG Analysis

Total CpG sites	CpG sites ($\geq 4X$)	CpG sites (Prob $\geq 50\%$)	CpG sites (Prob $\geq 70\%$)	CpG sites (Prob $\geq 90\%$)	CpG sites (Prob $\geq 95\%$)	CpG sites (Prob $\geq 99\%$)
1,000,000	1,000,000	10,000	5,000	2,500	50	0

Total CpG sites : Total CpG sites

CpG sites ($\geq 4X$) : CpG sites supported by 4 reads or more

CpG sites (Prob $\geq 50\%$) : CpG sites which have more than or equal to 50% methylated reads from supported reads

CpG sites (Prob $\geq 70\%$) : CpG sites which have more than or equal to 70% methylated reads from supported reads

CpG sites (Prob $\geq 90\%$) : CpG sites which have more than or equal to 90% methylated reads from supported reads

CpG sites (Prob $\geq 95\%$) : CpG sites which have more than or equal to 95% methylated reads from supported reads

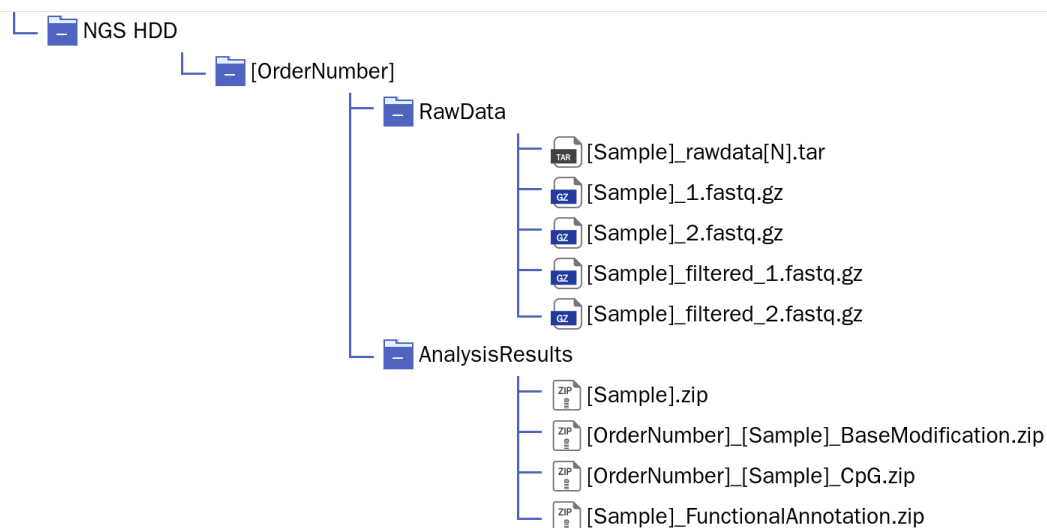
CpG sites (Prob $\geq 99\%$) : CpG sites which have more than or equal to 99% methylated reads from supported reads

Download List

Please make sure to check the integrity of the files. The sequencing files will be deleted from our server 3 months after the analysis report is released; Please contact us within 3 months if you encounter a problem with the data.

File Name	File Size (byte)	md5sum
SAMPLE1_rawdata1.tar	1.0GB	6f7e6b7a3b0a99f7e5782dc63d4887b9
SAMPLE1_1.fastq.gz	900.0MB	63f157a648a731eae5270f964f41f10
SAMPLE1_2.fastq.gz	920.0MB	8db0e61b6b72384751fd5853e5b4a900
SAMPLE1_filtered_1.fastq.gz	500.0MB	5bfe29083d0075c7e956791267314a01
SAMPLE1_filtered_2.fastq.gz	490.0MB	9d4a7b0f73cdd28b770b74dababd3f92
SAMPLE1.zip	20.0MB	7a5f65b16683c12c35ad23d65ed1d9b0
SAMPLE1_FunctionalAnnotation.zip	15.0MB	d7759432732369919aaad8795ec2a2c5
HN00000000_SAMPLE1_BaseModification.zip	5.0MB	6e14c1e63c93a83d9e24d29e43a477c7
HN00000000_SAMPLE1_CpG.zip	10.0MB	2a89b9a56e088a04852b377f7d8bcb41

 Folder
  Text file
  Excel file
  TAR
  GZIP
  ZIP
  File



FAQ

Q Why do I need to check the md5sum values, and how can I check it? (Windows system)

A NGS data tend to have a large files size which makes them more likely to be corrupted during file transfer. So it's important that you check the md5sum of the files after receiving them to make sure what you received are what we gave.

Checking md5 hash in a Windows system

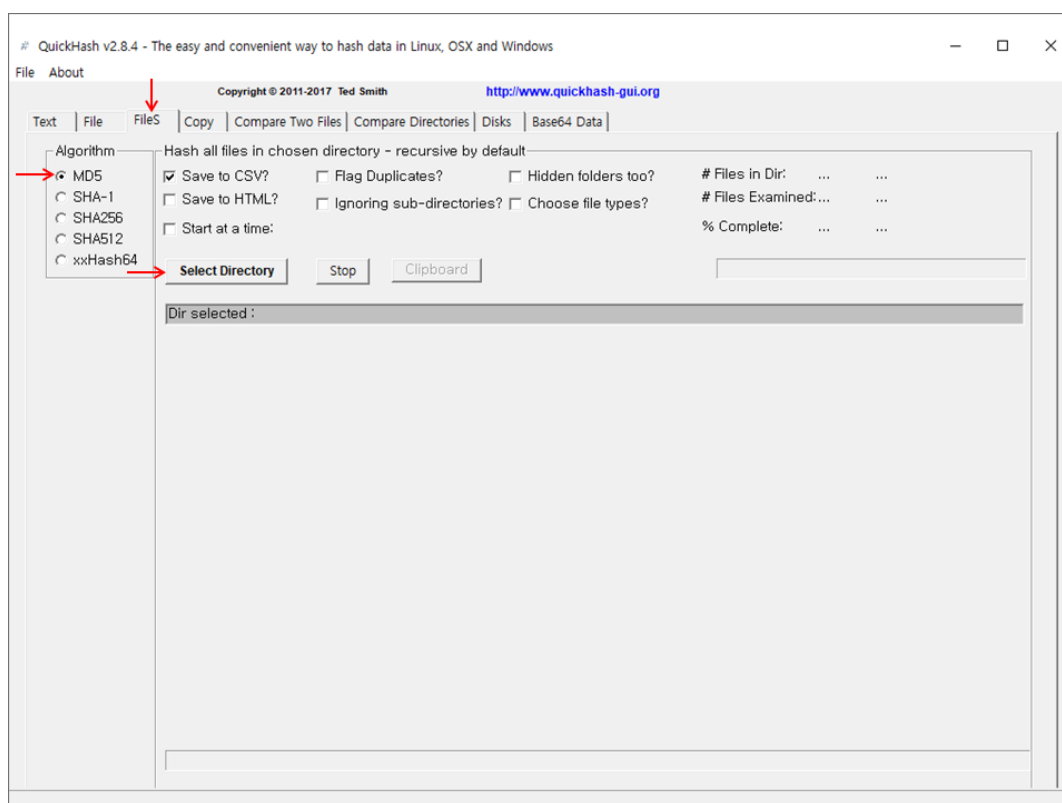
Windows does not provide a program for checking md5sum by default. An external program such as QuickHash-Windows can be used instead. (<https://www.quickhash-gui.org/downloads>)

STEP 1 Download QuickHash-Windows from the website, and unzip the file.

STEP 2 Take a look at the UserManual.pdf file inside the zip file, and execute the .exe file.

Quickhash-GUI.exe	2,090,414	6,505,472
sqlite3-win32.dll	429,646	852,754
sqlite3-win64.dll	717,149	1,742,848
UserManual.pdf	512,697	576,987

STEP 3 Click on the "FileS" tab, and select [MD5] as the Algorithm.



STEP 4 Click "Select Directory" and choose the directory where the files to be checked are located in. The output can be saved as a csv or txt file. The process may take some time depending on the performance of the system being used.

STEP 5 Compare the newly calculated md5 value with the md5 value provided to you through the Analysis Report.

Q Why do I need to check the md5sum values, and how can I check it? (Linux system)

A NGS data tend to have a large files size which makes them more likely to be corrupted during file transfer. So it's important that you check the md5sum of the files after receiving them to make sure what you received are what we gave.

Checking md5 hash in a Linux system

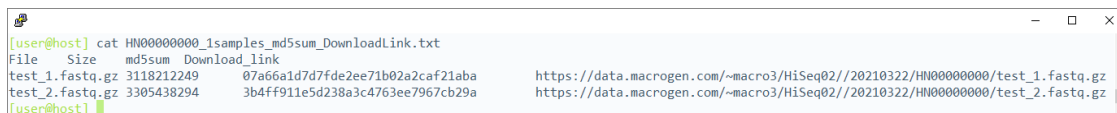
Linux systems have an internal md5sum program under /user/bin/md5sum.

md5sum has a "-c" option, which reads the md5sums from the input file and checks them simultaneously.

Usage: \$ md5sum -c [input file name]

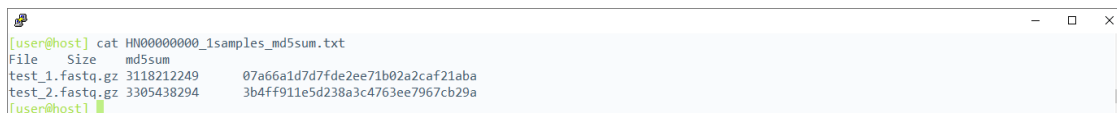
STEP 1 Macrogen provides a text file containing the md5sum of deliverables you'll be receiving, which you can use to validate the integrity of the files. You can download this file by clicking on the "md5sum List" button in the "Download List" page. The text file will have the following name and format depending on how you're receiving your data:

○ Via download link : <OrderNumber>_#samples_md5sum_DownloadLink.txt



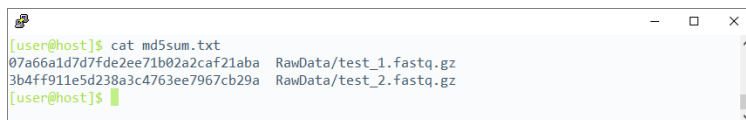
```
[user@host] cat HN00000000_1samples_md5sum_DownloadLink.txt
File      Size      md5sum      Download_link
test_1.fastq.gz 3118212249 07a66a1d7d7fde2ee71b02a2caf21aba https://data.macrogen.com/~macro3/HiSeq02//20210322/HN00000000/test_1.fastq.gz
test_2.fastq.gz 3305438294 3b4ff911e5d238a3c4763ee7967cb29a https://data.macrogen.com/~macro3/HiSeq02//20210322/HN00000000/test_2.fastq.gz
```

○ Via HDD : <OrderNumber>_#samples_md5sum.txt



```
[user@host] cat HN00000000_1samples_md5sum.txt
File      Size      md5sum
test_1.fastq.gz 3118212249 07a66a1d7d7fde2ee71b02a2caf21aba
test_2.fastq.gz 3305438294 3b4ff911e5d238a3c4763ee7967cb29a
```

○ You can also find "md5sum.txt" located inside the HDD delivered to you.



```
[user@host]$ cat md5sum.txt
07a66a1d7d7fde2ee71b02a2caf21aba RawData/test_1.fastq.gz
3b4ff911e5d238a3c4763ee7967cb29a RawData/test_2.fastq.gz
```

STEP 2 Use "md5sum -c" to validate the integrity of the file you've received. The input file for md5sum -c has to be delimited by two spaces with the md5sum column appearing before the file name, just like the sample image of "md5sum.txt" file shown above. As you can see, the two other files above are not formatted this way and need to be altered to be used as input for md5sum -c. You can manually exclude the header and cut out "File" and "md5sum" column from the files, or simply run the following command:

\$ awk '{print \$3 " " \$1}' <md5sum_file> | grep -v File

STEP 3 "md5sum -c" reads the input containing the md5 value of a file, and checks whether the md5 value of that file matches what's written inside the input file. This action outputs "OK" if the md5 value matches, and "FAILED" if otherwise. Check if the command outputs "OK" for all the files. (Refer to image below)



```
user@host
[user@host] awk '{print $3 " " $1}' HN00000000_1samples_md5sum_DownloadLink.txt | grep -v File > md5sum.txt
[user@host] cat md5sum.txt
07a66a1d7d7fde2ee71b02a2caf21aba test_1.fastq.gz
3b4ff911e5d238a3c4763ee7967cb29a test_2.fastq.gz
[user@host] md5sum -c md5sum.txt
test_1.fastq.gz: OK
test_2.fastq.gz: OK
```

Q I want to see the data produced by MacroGen. How can I open the files?

A NGS data tend to have large file sizes, and are not user-friendly to work with in a Windows environment. We recommend that you use Linux system for smoother operation.

Q Where can I find information for Illumina adapter sequences?

A Information on Illumina adapters can be found in this support document:
Adapter Sequences Intro
(<https://support-docs.illumina.com/SHARE/AdapterSequences/Content/SHARE/FrontPages/AdapterSeq.htm>)

Deliverables List

Raw Data

File Type	File Name	Description
fastq	[Sample]_1.fastq.gz	Raw read1 sequence data
fastq	[Sample]_2.fastq.gz	Raw read2 sequence data
fastq	[Sample]_HiFi.fastq.gz	HiFi data in fastq format
bam	[Sample].hifi_reads.bam	HiFi data in binary format

De novo Assembly Results

File Type	File Name	Description
Contig	*contigs.fasta	Sequences of assembled contigs.
BLAST	*[Sample]_BLAST.xlsx	BLAST analysis results against NCBI NT database.

Annotation Results

File Type	File Name	Description
Genome	*.fna	Whole nucleotide sequence.
Genome	*.fsa	Whole nucleotide sequence (Detailed description).
Gene Info.	*.gff	GFF3 format. https://github.com/The-Sequence-Ontology/Specifications/blob/master/gff3.md
Gene Info.	*.ffn	Fasta format of transcript sequence.
Gene Info.	*.faa	Fasta format of amino acid.
Annotation	*_annotation.xlsx	Excel file that saves contents written in HTML.
Annotation	*_EggNOG.xlsx	Excel file containing the results of the analysis of EggNOG database.
NCBI Format	*.sqn	NCBI's Sequin format (Edited with Sequin).
NCBI Format	*.gbk	GenBank format.
NCBI Format	*.tbl	Tap separation formality of NCBI.

File Format

● FASTQ File

FASTQ Format

Example:

FASTQ file consists of four lines.
 Quality score is represented with each character.
 One character matches its base with Phred+33

Line 1 : Sequence identifier

Line 2 : Nucleotide sequences

Line 3 : Quality score identifier line - character '+'

Line 4 : Quality score

```

@A00125:17:H2HFJDMXX:1:1101:3170:1000 1:N:0:ATGCCTAA
GAAACACGATGACACTCACATGGCACTCACATTTCACTCCTTTCTAAGTGATTGCAATATTAATTCATATTTAATATT
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
@A00125:17:H2HFJDMXX:1:1101:9408:1000 1:N:0:ATGCCTAA
TGTGCGAAGGAAATCATTTCAGATGACAGTGTTAACCATGGTCAAAGGACCATTCTGCTCTATCCTTCTTAGAAGCTTCC
+
FFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF

```

Phred Scores

Phred quality score numerically expresses the accuracy of each nucleotide. Higher Q number signifies higher accuracy. For example, if Phred assigns a quality score of 30 to a base, the chances of having base call error are 1 in 1000.

Phred Quality Score	Probability of Incorrect Base Call	Base Call Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%
60	1 in 1,000,000	99.9999%

$Q = -10 \log_{10}(\text{error rate})$ Encoding : ASCII Character Code = Phred Quality Value + 33

Q-Score Binning

The Illumina's quality score (Q score) uses quality binning. Quality scores have been specially calibrated for Illumina systems and their consumables. The quality score table below is typically updated when significant characteristics of the sequencing platform change, such as new hardware, software, or chemistry versions.

Phred Quality Score	Example of Empirically Mapped Q-Scores
N(no call)	N(no call)
2-9	7
10-19	11
20-24	22
25-29	27
30-34	32
35-39	37
40-45	42

BAM File

BAM

The BAM is a compressed binary format of a SAM (Sequence Alignment Map). The BAM file contains information about sequence alignment of reads against a large reference sequence.

Example:

```

@HD VN:1.5 SO:coordinate
@SQ SN:chrM LN:16569
@SQ SN:chr1 LN:248956422
@SQ SN:chr2 LN:242193529
@SQ SN:chr3 LN:198295559
@SQ SN:chr4 LN:190214555
@SQ SN:chr5 LN:181538259
@SQ SN:chr6 LN:170805979
@SQ SN:chr7 LN:159345973
@SQ SN:chr8 LN:145138636
@SQ SN:chr9 LN:138394717
@SQ SN:chr10 LN:133797422
@SQ SN:chr11 LN:135086622
@SQ SN:chr12 LN:133275309
@SQ SN:chr13 LN:114364328
@SQ SN:chr14 LN:107043718
@SQ SN:chr15 LN:101991189
@SQ SN:chr16 LN:90338345
@SQ SN:chr17 LN:83257441
@SQ SN:chr18 LN:80373285
@SQ SN:chr19 LN:58617616
@SQ SN:chr20 LN:64444167
@SQ SN:chr21 LN:46709983
@SQ SN:chr22 LN:50818468
@SQ SN:chrX LN:156040895
@RG SN:chrY LN:57227415
@PG ID:A00125 PL:Illumina SM:Sample LB:SS6
@PG ID:bwa PN:bwa VN:0.7.17-r1188 CL:bwa mem
@PG ID:MarkDuplicates VN:2.18.2-SNAPSHOT CL:MarkDuplicates PN:MarkDuplicates
@PG ID:GATK ApplyBQSR VN:4.0.5.1 CL:ApplyBQSR PN:GATK ApplyBQSR

A00125:511:HFVLSDSX2:3:1167:10176:32142 83 chr1 826845 40 101M = 826766
-180 CCCACCTCGACATCCACAGCGAGGCAATGAAGAAGCCCTGCCAAGGAAGAGCCCGCTTCTCAGTGGGACACCGGAAGGTAGACACCAACAGTCACCGC EEH<FG@<C<H=@FH<H
<FAD7FFI<>7D<7D<7GHHHI@GHJ=@GE=@E@GHHBGICBIBJ@F@GGE>J>HBGGE=@GF7@E>J>HHJ=>J@FBJ>CHB XA:Z:chr8:+232954,101M,1: MC:Z:101M MD:Z:48G52
PG:Z:MarkDuplicates RG:Z:A00125 NM:i:1 AS:i:96 XS:i:96
A00125:511:HFVLSDSX2:3:1650:26829:19492 83 chr1 826845 27 101M = 826807
-139 CCCACCTCGACATCCACAGCGAGGCAATGAAGAAGCCCTGCCAAGGAAGAGCCCGCTTCTCAGTGGGACACCGGAAGGTAGACACCAACAGTCACCGC EEH<FG@<C<H=@FH<H
>FAD7FFI<>7D<7D<7GHHHI@GHJ=@GE=@E@GHHBGICBIBJ@F@GGE>J>HBGGE=@GF7@E>J>HHJ=>J@FBJ>CHB XA:Z:chr8:+232954,101M,1: MC:Z:101M MD:Z:48G52
PG:Z:MarkDuplicates RG:Z:A00125 NM:i:1 AS:i:96 XS:i:96

```

Header section

Alignment section

More information about BAM format can be found here:

<https://samtools.github.io/hts-specs/SAMv1.pdf>

Header Line

Tag	Description
@HD	The header line
@PG	Program and command line
@RG	Read group. platform, sample name information
@SQ	Reference sequence dictionary. The order of @SQ lines defines the alignment sorting order.
@CO	One-line text comment. Unordered multiple @CO lines are allowed.

SAM file : Alignments section mandatory fields

Field	Regexp / Range	Description
QNAME	[!-?A~]{1,255}	Query template NAME
FLAG	[0,2^16-1]	Bitwise FLAG
RNAME	\ !-()-+---~][!-~]*	Reference sequence NAME
POS	[0,2^31-1]	1-based leftmost mapping POSition
MAPQ	[0,2^8-1]	MAPping Quality
CIGAR	\ ([0-9]+[MIDNSHPX=])+	CIGAR string
RNEXT	\ = !-()-+---~][!-~]*	Ref. name of the mate/NEXT read
PNEXT	[0,2^31-1]	Position of the mate/NEXT read
TLEN	[-2^31+1,2^31-1]	Observed Template LENgth
SEQ	\ [A-Za-z=.]+	segment SEQUENCE
QUAL	[!-~]+	ASCII of Phred-scaled base QUALity+33

Program Used in Analysis

Software Version

Software	Version
FastQC	0.11.7
Trimmomatic	0.38
Jellyfish	1.1.12
GenomeScope	-
Microbial Genome Analysis	SMRTLINK_13.0.0.207600
Inspector	1.0.1
ipdSummary	SMRTLINK_13.0.0.207600
motifMaker	SMRTLINK_13.0.0.207600
pb-CpG-tools	1.1.0
Pilon	1.22
Prokka	1.14.6
eggNOG	4.5
InterProScan	5.34-73.0
BWA	0.7.17-r1188
pbmm2	12.1
BUSCO	5.1.3
BLAST	2.14.0
pyani	0.2.7
Circos	0.69.9

FastQC

0.11.7

FastQC 0.11.7 aims to provide a simple way to do some quality control checks on raw sequence data coming from high throughput sequencing pipelines. It provides a modular set of analyses which you can use to give a quick impression of whether your data has any problems of which you should be aware before doing any further analysis.

More information can be found here:

<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>

Citation

<http://www.bioinformatics.babraham.ac.uk/projects/fastqc>

Trimmomatic

0.38

Incomplete removal of adapter sequences from NGS data can ultimately affect the accuracy of analysis. To avoid this, Trimmomatic 0.38 is used to remove adapter sequences. Depending on the library type used for data production, appropriate adapter sequence is used to remove the said sequences from the data. In addition to removing adapter sequences, Trimmomatic trims out bases of low quality.

More information can be found here:

<http://www.usadellab.org/cms/?page=trimmomatic>

Citation

<https://pubmed.ncbi.nlm.nih.gov/24695404/>

Jellyfish

1.1.12

Jellyfish 1.1.12 is a program that counts k-mers in DNA. It provides information that can be used in many analyses of DNA sequences such as genome size prediction, genome coverage confirmation and repeat sequence ratio calculation by counting the abundance of a particular k-mer in the sequence.

More information can be found here:

<http://www.genome.umd.edu/jellyfish.html>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3051319/>

GenomeScope

GenomeScope can infer the global properties of a genome from unassembled sequenced data. GenomeScope uses the k-mer count distribution and within seconds produces a report and several informative plots describing the genome properties.

More information can be found here:

<http://qb.cshl.edu/genomescope>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5870704/>

Microbial Genome Analysis

SMRTLINK_13.0.0.207600

Microbial Genome Analysis Application SMRTLINK_13.0.0.207600 is a tool for assembly of PacBio microbial genome data from SMRTlink. It has advantages to de novo assembly of small prokaryotic genomes between 1.9~10Mb and companion plasmids between 2~220 kb. The Microbial Genome Assembly can check whether contigs are circular form or not and rotate contig start sites as origin of replication for each contigs automatically.

More information can be found here:

<https://www.pacb.com/wp-content/uploads/SMRT-Link-User-Guide-v13.0.pdf>

Citation

-

Inspector

1.0.1

Inspector, which reports exact locations, sizes, and types of assembly errors without being affected by genetic variants. In addition, Inspector improves assembly results by correcting discovered errors.

More information can be found here:

<https://github.com/Maggi-Chen/Inspector>

Citation

<https://genomebiology.biomedcentral.com/articles/10.1186/s13059-021-02527-4>

ipdSummary

SMRTLINK_13.0.0.207600

The ipdSummary detects DNA base-modifications from the kinetic signatures of the aligned BAM file.

More information can be found here:

<https://github.com/PacificBiosciences/kineticsTool>

Citation

-

motifMaker

SMRTLINK_13.0.0.207600

The motifMaker discovers DNA methylation motifs from methylated DNA sequences in prokaryotic genomes.

More information can be found here:

<https://github.com/wangdi2014/MotifMaker>

Citation

-

pb-CpG-tools

1.1.0

pb-CpG-tools 1.1.0 can analyze CpG/5mC data from alignments between PacBio HiFi reads containing CpG information generated by Jasmine and reference genome.

More information can be found here:

<https://github.com/PacificBiosciences/pb-CpG-tools>

Citation

-

Pilon

1.22

Pilon 1.22 is a software tool which can be used to automatically improve draft assemblies. It significantly improves draft genome assemblies by correcting bases, fixing mis-assemblies and filling gaps

More information can be found here:

<https://github.com/broadinstitute/pilon/wiki>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4237348/>

Prokka

1.14.6

Prokka 1.14.6 is a pipeline that performs series of process automatically. At the end of the pipeline, It gives GBK files as well as various types of files such as GFF3, SQN.

More information can be found here:

<http://www.vicbioinformatics.com/software.prokka.shtml>

Citation

<https://academic.oup.com/bioinformatics/article/30/14/2068/2390517>

eggNOG

4.5

EggNOG (Evolutionary genealogy of genes: Non-supervised Orthologous Groups) is a database of orthology relationships, functional annotation, and gene evolutionary histories. It incorporates data from bacteria (COGs), archaea (arCOGs), eukaryotes (KOGs), and viruses (viral OGs).

More information can be found here:

<http://eggnog.embl.de>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4702882/>

InterProScan

5.34-73.0

InterProScan is a protein functional analysis tool that combines a variety of protein signature databases to identify domains and motifs in protein sequences.

More information can be found here:

<https://www.ebi.ac.uk/interpro/about/interproscan/>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3998142/>

BWA

0.7.17-r1188

BWA is a software package for mapping DNA sequences against a large reference genome, such as the human genome

More information can be found here:

<https://github.com/lh3/bwa>

Citation

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2705234/>

pbmm2

12.1

pbmm2 12.1 is a Minimap2 SMRT wrapper for PacBio data. Minimap2 is a sequence alignment program that can find overlaps between long reads, or map reads to a reference genome, align assembly-to-assembly.

More information can be found here:

<https://github.com/PacificBiosciences/pbmm2>

Citation

-

BUSCO

5.1.3

The Benchmarking Universal Single-Copy Orthologous (BUSCO, 5.1.3) can assess assembly quality by comparison between predicted genes from genome assembly and near-universal single-copy orthologs DB. If assembly display higher complete BUSCOs, the assembly could be regarded as assembly with good quality.

More information can be found here:

<http://busco.ezlab.org/>

Citation

<https://academic.oup.com/bioinformatics/article/31/19/3210/211866>

BLAST

2.14.0

The Basic Local Alignment Search Tool (BLAST, 2.14.0) finds regions of local similarity between sequences. The program compares nucleotide or protein sequences to sequence databases and calculates the statistical significance of matches. BLAST can be used to infer functional and evolutionary relationships between sequences as well as help identify members of gene families

More information can be found here:

<https://blast.ncbi.nlm.nih.gov/Blast.cgi>

Citation

<https://pubmed.ncbi.nlm.nih.gov/2231712/>

pyani

0.2.7

pyani is an application for the whole genome classification of microbes using Average Nucleotide Identity (ANI). ANI is a measure of similarity between two genomes. It is calculated by comparing the nucleotides in the genome. The higher the ANI, the more similar the genomes are.

More information can be found here:

<https://github.com/widdowquinn/pyani>

Citation

<https://pubs.rsc.org/en/content/articlelanding/2016/ay/c5ay02550h>

Circos

0.69.9

Annotated map was drawn by Circos 0.69.9

More information can be found here:

<https://circos.ca/>

Citation

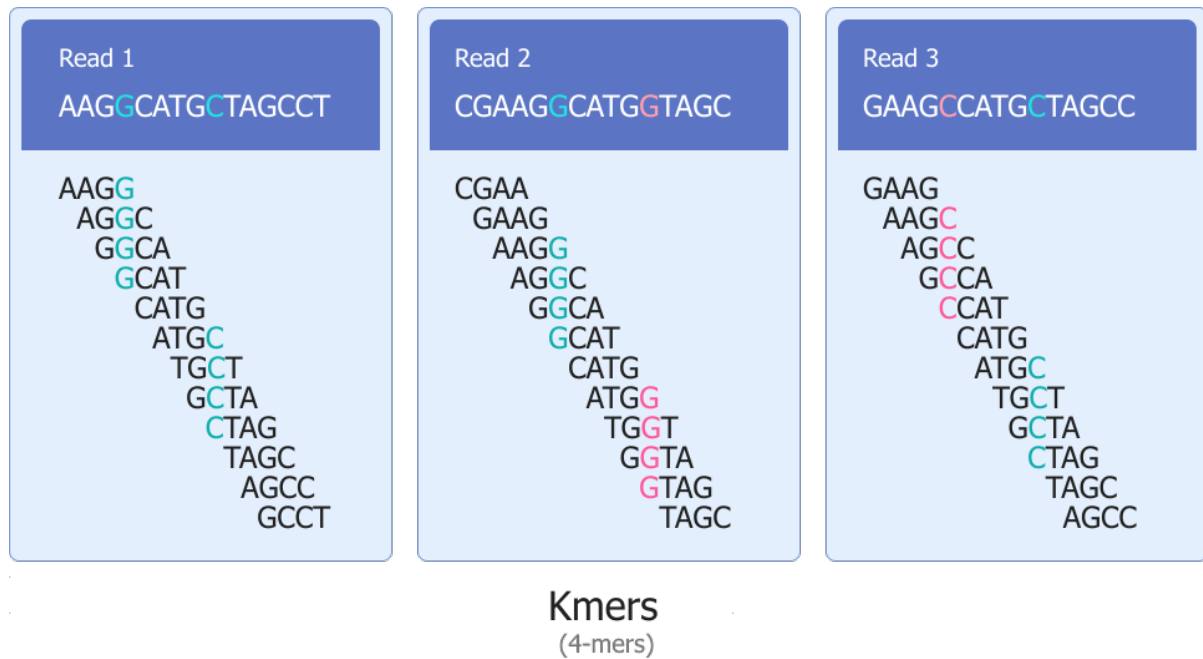
<https://genome.cshlp.org/content/19/9/1639>

Report Reading Guide in Korean

K-mer Analysis

유전체학에서 K-mer란 특정 서열 내에 K 값의 길이를 가지는 부분 서열의 집합을 의미하며, K-mer를 count 하는 방법을 이용해서 유전체의 대략적인 크기 및 특성(Heterozygosity 등)을 in silico 적으로 추정할 수 있습니다.

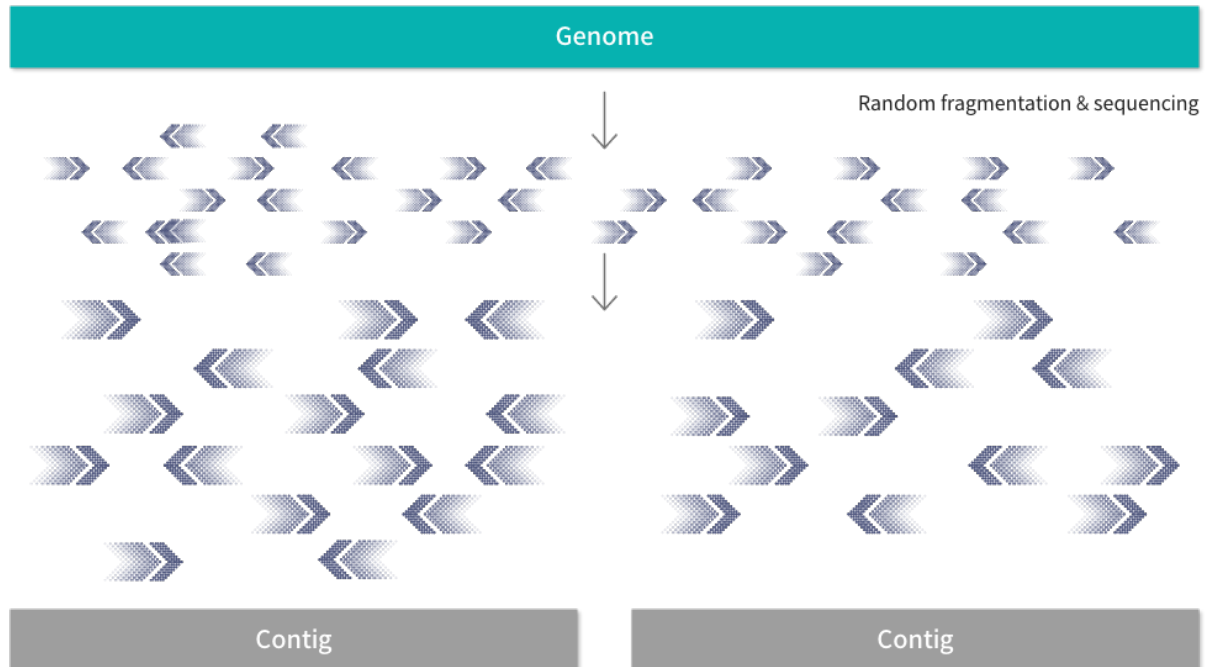
Example of K-mer



K-mer 분석은 적절한 시퀀싱 생산량을 결정하거나, de novo assembly로 만들어진 contig가 잘 만들어졌는지 확인하는 데 도움을 줍니다. 저희는 기존에 연구된 target species의 정보가 있을 경우 이를 K-mer 분석 결과와 비교하여 분석 시료가 잘 일치하는지를 확인하고, de novo assembly 완료 이후에는 total contig base와 비교하여 유전체 조립의 완성도를 가능하고 있습니다.

De novo Assembly

Schematic Workflow of De novo Assembly

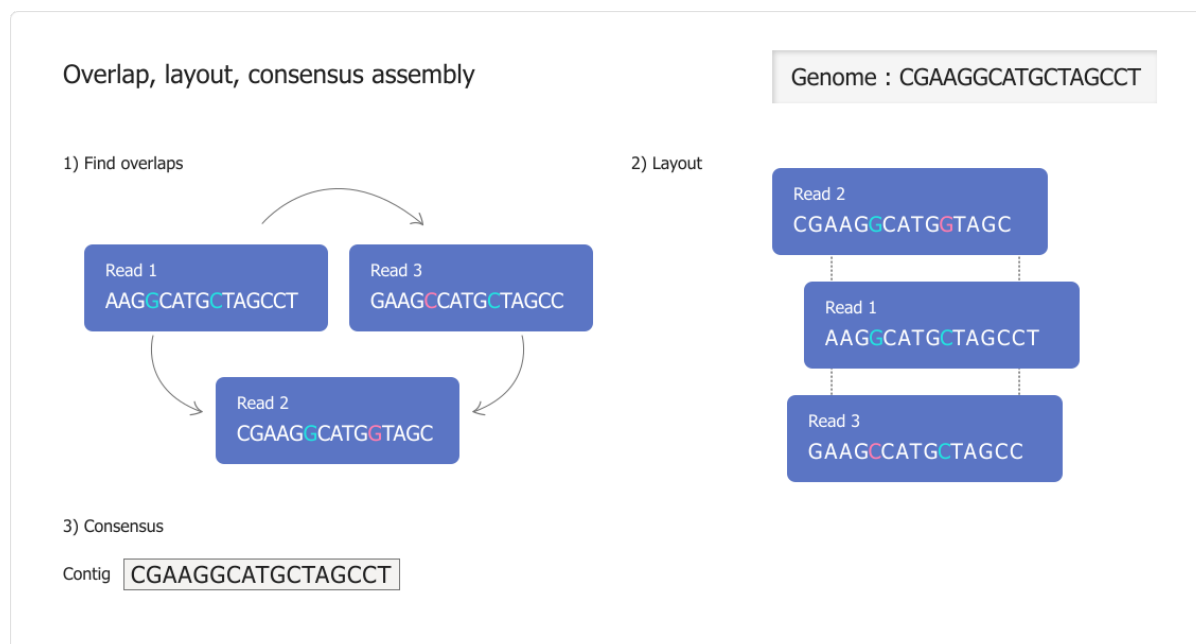


De novo assembly는 'from the beginning', 'anew'의 뜻을 나타내는 라틴어 표현, 'de novo'에서 짐작할 수 있듯이 참조할 수 있는 다른 reference의 도움 없이 DNA 서열 정보를 이용하여 처음부터 bottom-up 방식으로 유전체 초안 지도를 만드는 것을 의미합니다. De novo assembly는 지금까지 밝혀지지 않은 미생물이나 동, 식물 등 새로운 생물종의 표준 유전체 정보를 알아내기 위해 진행하며, 그 결과는 유전자 예측과 annotation, 유전체 구조 변이 탐색 등 DNA 서열을 이용한 모든 연구의 근간이 됩니다.

Sequencing platform마다 데이터(sequencing reads)를 생산하는 방식이 다르고, read의 길이나 특성에 따라 de novo assembly를 하는 방법은 차이가 있지만, 일반적으로는 whole genome DNA를 짧게 잘라 DNA 단편을 만들고, 각각의 DNA 단편을 sequencing 한 후 이를 조립하여 유전체 서열을 작성하는 방식으로 진행됩니다.

Sequencing 관련 기술이 비약적으로 발전하였음에도 NGS로 생산하는 read의 길이가 whole genome에 비해 매우 짧고, 생산 비용의 문제 등 여러 한계가 존재하기 때문에, 데이터의 특성과 양을 고려하여 assembler를 선택하는 것이 중요합니다.

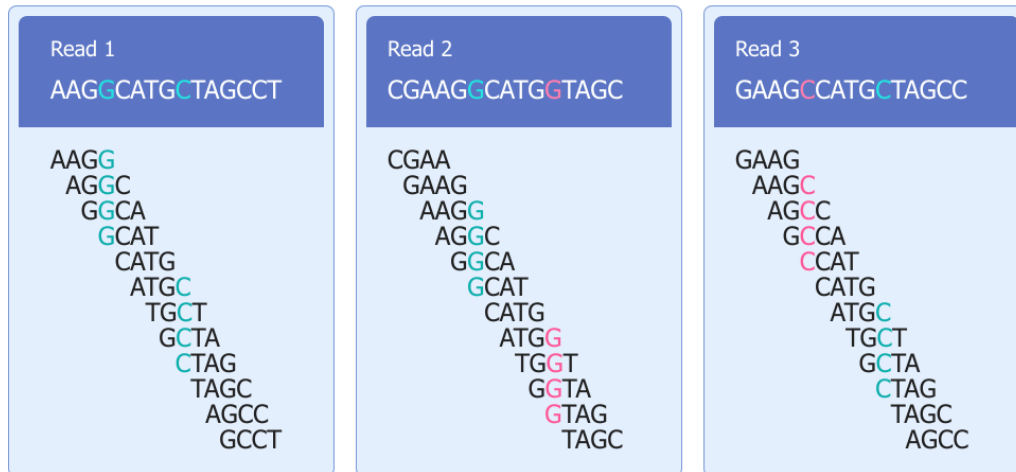
Algorithms of De novo Assembly



De Bruijn graph assembly

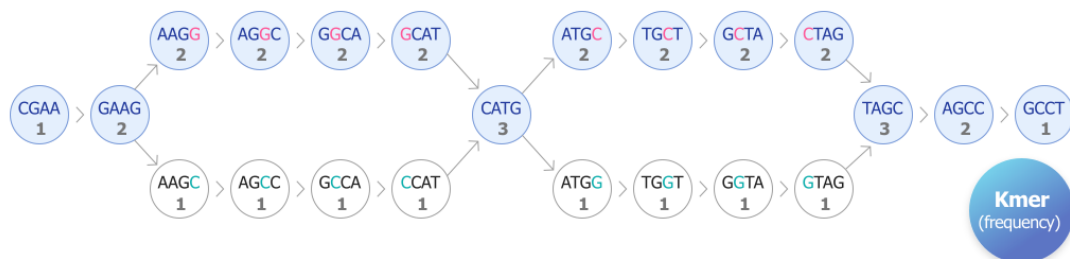
Genome : CGAAGGCATGCTAGCCT

1) Make kmers

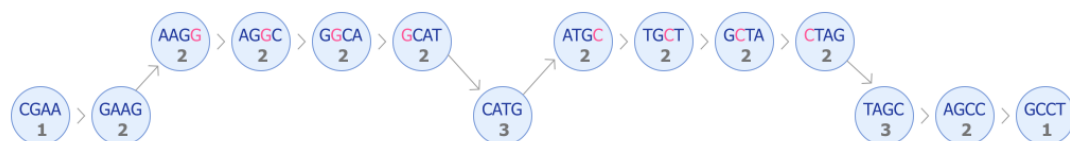


Kmers
(4-mers)

2) Build graph



3) Walk graph and output contigs



Contig **CGAAGGCATGCTAGCCT**

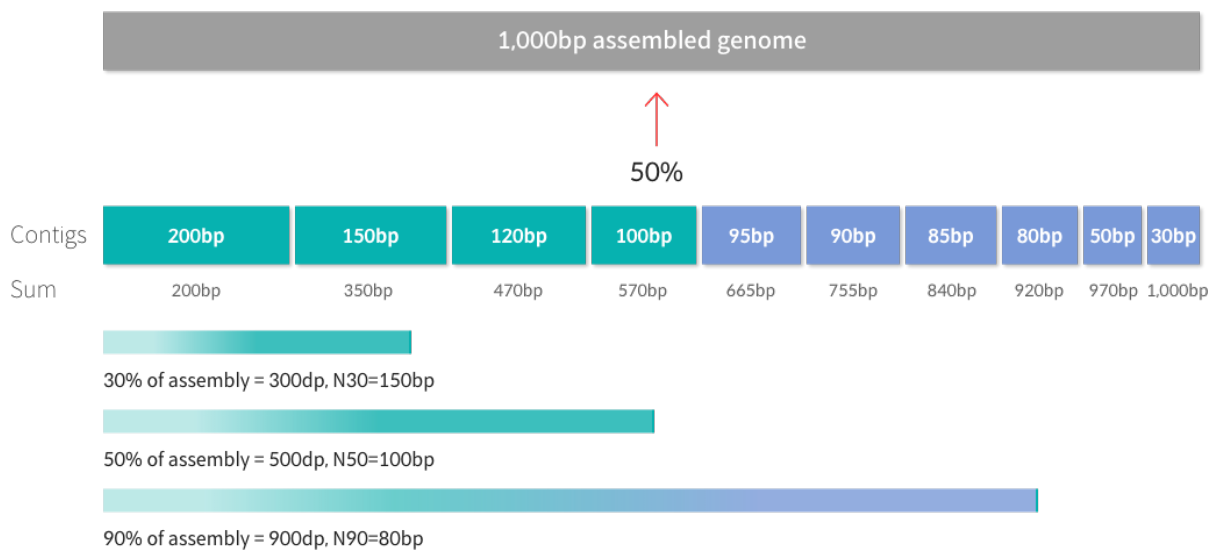
조각난 서열을 조립하는 방식은 각 assembler의 알고리즘에 따라 다르지만 대체로 read 간의 겹치는 서열 정보를 확인하며 확장하는 Overlap-Layout-Consensus 방법과 각 read를 정해진 K 값의 길이를 가지는 조각(K-mer)으로 나누고, 이로부터 1base씩 연장하여 가장 기댓값이 높은 서열을 선택하는 de Bruijn graph 방법 중 하나를 채택하고 있습니다.

Illumina platform으로 생산된 짧은 read의 경우 컴퓨팅 자원과 assembly 효율 면에서 강점을 보이는 de Bruijn graph 방식을 주로 사용하며, 해당 알고리즘을 사용하는 assembler에는 SPAdes, Platanus, SOAPdenovo 등이 있습니다. De Bruijn graph 방식의 assembler는 K 값에 따라서 결과물이 달라지기 때문에, 여러 K 값으로 de novo assembly를 진행한 후 contig 개수, total contig bases, N50 등을 고려하여 가장 결과가 좋다고 판단되는 결과물을 제공하고 있습니다.

Contig

Contig는 본래 'set of overlapping DNA segments'를 총칭하나, de novo assembly에 한정할 경우 sequencing platform에서 생산한 read 간의 겹치는 정보를 기반으로 이들을 연결하여 조립한 서열을 의미하며, 별도로 scaffolding(mate-pair library와 같은 insert size가 긴 서열을 이용하여 두 개 이상의 contig를 gap으로 연결하는 작업)을 수행하지 않는다면 de novo assembly는 이 단계에서 완성되기 때문에, contig는 de novo assembly 과정에서 생산된 중간, 혹은 최종 결과물이라고 할 수 있습니다. Sequencing platform, 유전체별 특성에 따라 de novo assembly의 결과물이 달라지기에 절대적인 기준은 존재하지 않으나 몇몇 수치를 통해 de novo assembly가 잘 이루어졌는지 가능해 볼 수 있습니다. 조립한 contig의 수가 target 유전체의 chromosome 수에 근접할수록, total contig bases가 target 유전체의 크기 (혹은 데이터를 통해 예측된 유전체 크기)에 가까울수록, contig의 max length가 길수록 분석 결과가 target 유전체의 whole genome을 잘 반영하고 있다고 할 수 있습니다.

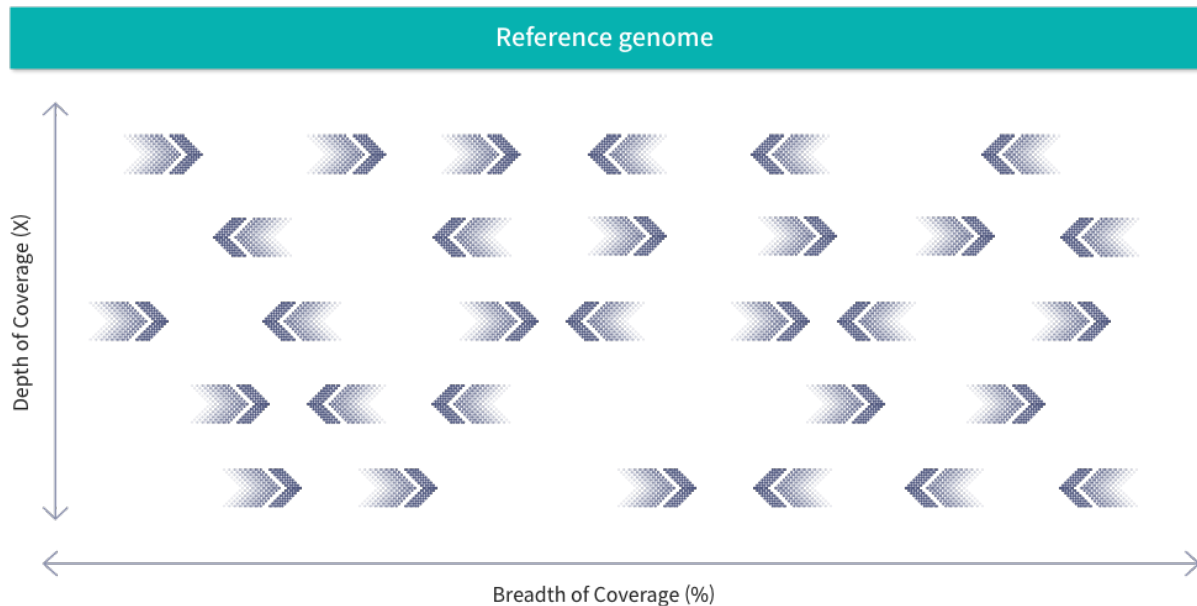
Schematic Explanation of N50 (Nx)



그 외에도 N50(or Nx)를 통해서도 de novo assembly의 완성도를 추측할 수 있습니다. N50이란 완성된 contig들을 길이 순으로 나열하여 길이가 긴 것부터 순차적으로 더했을 때 그 값이 전체 contig 길이의 50%(Nx의 경우 x%) 이상을 만족하는 지점에서의 contig의 길이를 의미하며 해당 수치가 클수록 길이가 긴 contig가 주로 형성된 것으로 추정할 수 있습니다.

Self-Mapping Analysis

Assembly 결과를 평가하기 위해 유전체 assembly에 사용되었던 short read sequence를 de novo assembly 결과(contig 혹은 scaffold)에 mapping 하여 다음과 같은 항목을 계산합니다.



Depth of Coverage

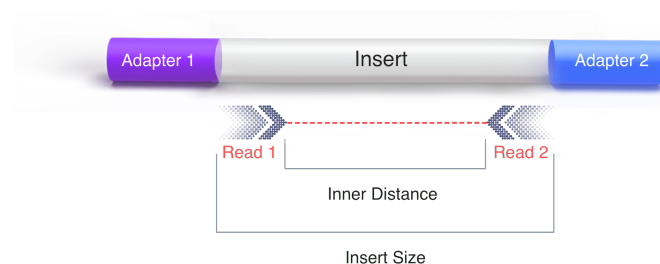
Depth of coverage (이하 depth로 표기)는 reference의 특정 염기 위치에, 몇 개의 read가 mapping이 되었는지를 의미합니다. 예를 들어 100X는 특정 염기 위치에 리드가 100번 반복하여 시퀀싱되었다는 것을 나타내며, average depth가 크다는 것은 조립된 유전체 대비 data가 많이 생산되었음을 의미합니다. 일반적으로 de novo assembly를 위해서는 유전체 크기 대비 80-150X 정도의 시퀀싱 데이터 생산을 권장하고 있습니다. 이는 80-150X 정도의 depth가 충족되었을 때, 시퀀싱 에러로 인해 발생하는 de novo assembly 에러를 방지하여 분석에서 높은 정확도와 신뢰도를 확보할 수 있기 때문입니다. 하지만, 200X 이상의 지나치게 높은 depth가 충족되게 되면, 역으로 시퀀싱 에러 base를 구분할 수 없게 되기 때문에 de novo assembly에 부정적인 영향을 미치기도 합니다.

Breadth of coverage

Breadth of coverage(이하 coverage로 표기)는 mapping이 이루어진 short read가 전체 de novo assembly 결과를 어느 정도 커버하고 있는지를 나타낸 수치로, coverage 값이 클수록 de novo assembly 결과의 더 많은 영역이 매치되었다는 의미입니다. Depth가 1X 이상인 영역을 모두 더한 결괏값을 de novo assembly 결과와 비교하여 %로 계산하고 있습니다.

Insert Size

Insert란 시퀀싱이 이루어진 라이브러리에서 어댑터 서열을 제거한 것으로 paired-end read(R1, R2)와 그 사이의 inner mate 영역으로 구성되어 있습니다. 본 분석에서는 mapping 결과로부터 Insert의 크기를 계산한 후 라이브러리 제작에 사용한 DNA 단편의 길이와 비교하였습니다. De novo assembly가 잘 이루어졌다면, insert size가 DNA 단편 크기와 유사할 것이며 이를 근거로 de novo assembly의 완성도를 판단할 수 있습니다. 일반적인 경우 paired-end 라이브러리의 insert는 350-450bp 사이의 평균값을 가지며, 표준 정규분포의 형태로 분포하게 됩니다.



BUSCO Analysis

De novo assembly 분석 결과를 평가하기 위해 BUSCO (Benchmarking sets of Universal Single-Copy Orthologs) 분석을 진행하였습니다. BUSCO 분석은 de novo assembly 결과에서 in silico 방법으로 유전자를 예측한 후 특정 taxon level에서 필수적으로 존재해야 하는 유전자가 예측된 유전자 내에 얼마나 존재하는지를 orthoDB에 근거하여 판단하는 방법입니다.

In silico 방법으로 예측한 유전자와 orthoDB 상의 유전자의 유사도를 통해 유전자를 분류하며, orthoDB 유전자 별로 하나의 예측 유전자와 거의 일치하면 'Complete and single BUSCOs', 복수의 예측 유전자와 거의 일치하면 'Complete and duplicated BUSCOs', 예측 유전자와 일부 서열만 비슷하면 'Fragmented BUSCOs'로 분류하며, orthoDB 상에는 존재하는 유전자가 예측 유전자에서 발견되지 않으면 'Missing BUSCOs'로 분류합니다.

Complete BUSCOs 수치가 높을수록 de novo assembly가 잘 되었다고 판단하며, 일반적으로는 90% 이상의 Complete BUSCOs 수치를 나타내면 de novo assembly 결과가 좋다고 판단하고 있습니다. 다만, 모델 생물과 유전적 거리가 먼 일부 종의 경우 de novo assembly quality와 관계없이 Complete BUSCOs 수치가 낮을 수 있습니다.

BUSCOs 분석은 bacteria level 또는 eukaryota level에 대해서만 결과를 제공해 드리고 있습니다.

Gene Prediction and Annotation

유전자 예측 및 annotation은 완성된 유전체 지도를 기반으로 단백질을 코딩하는 유전자와 non-coding RNA는 물론 조절 영역과 같이 functional element로서의 특성을 지니는 서열을 확인하고, 해당 서열이 어떠한 생물학적 기능과 연관되어 있는지 예측하는 것을 의미합니다.

일반적으로 원핵생물의 유전자 예측에는 Prokka를 사용합니다. Prokka는 ISfinder, NCBI Bacterial Antimicrobial Resistance Reference Gene DB, UniProtKB DB, HMM DB를 바탕으로 분석을 진행하며, 아래의 툴을 이용하여 각 functional element를 예측합니다.

Software	Role
Prodigal	CDS prediction
RNAmmer	rRNA prediction
Aragorn	tRNA/tmRNA prediction
SignalP	Signal leader peptide prediction
Infernal	Non-coding RNA prediction

Prokka를 통해 예측된 유전자는 InterProScan과 EggNOG DB를 통해 functional annotation을 진행하게 됩니다. InterProScan은 단백질을 families level로 분류한 이후 DB에 등록된 도메인(Pfam, CDD, TIGRFAM 등)을 대상으로 분석을 진행하여 Prokka에서 예측된 유전자가 어떤 도메인을 가지고 있는지를 예측해 줍니다. EMBL에서 제공하는 EggNOG DB는 COG DB와 유사한 형태로 유전자를 기능에 따라 분류하고 그들의 자세한 functional annotation을 포함하고 있는 데이터 베이스입니다. 저희는 psi-BLAST 를 통해서 protein 서열을 EggNOG DB에 매치하여 Prokka에서 예측된 유전자의 기능을 annotation 하고 있습니다.



HEADQUARTER

MacroGen Gangnam HQ

Business & Support Center

MacroGen Bldg, 238, Teheran-ro,
Gangnam-gu, Seoul, Republic of Korea
Tel: +82-2-2180-7000
Web: www.macrogen.com
LIMS: dna.macrogen.com

MacroGen Genome Center

Laboratory & IT Center

[08511] 1001, 10F, 254, Beotkkot-ro,
Geumcheon-gu, Seoul, Republic of Korea
(Gasan-dong, World Meridian 1)
Tel: +82-2-2180-7000
Email1: ngs@macrogen.com(Overseas)
Email2: ngskr@macrogen.com
(Republic of Korea)
Web: www.macrogen.com
LIMS: dna.macrogen.com

SUBSIDIARY

MacroGen Europe

Laboratory, Business & Support Center

Meibergdreef 57, 1105 BA, Amsterdam,
the Netherlands
Tel: +31-20-333-7563
Email: ngs@macrogen.eu

Psomagen (MacroGen USA)

Laboratory, Business & Support Center

1330 Piccard Drive, Suite 103, Rockville,
MD 20850, United States
Tel: +1-301-251-1007
Email: inquiry@psomagen.com

MacroGen Singapore

Laboratory, Business & Support Center

3 Biopolis Drive #05-18, Synapse,
Singapore 138623
Tel: +65-6339-0927
Email: info-sg@macrogen.com

MacroGen Japan

Laboratory, Business & Support Center

16F Time24 Building, 2-4-32 Aomi,
Koto-ku, Tokyo 135-0064 JAPAN
Tel: +81-3-5962-1124
Email: ngs@macrogen-japan.co.jp

BRANCH

MacroGen Spain

Laboratory, Business & Support Center

Av. Sur del Aeropuerto de Barajas,
28. Office B-2, 28042 Madrid, Spain
Tel: +34-911-138-378
Email: info-spain@macrogen.com

MacroGen Belgium

Laboratory, Business & Support Center

Oxfordlaan 70, 6229 EV Maastricht,
Netherlands
Tel: +31-20-333-7563
Email: info.be@macrogen.eu

MacroGen Italy

Laboratory, Business & Support Center

Viale Ortles, 22/4, 20139 Milano,
MI, Italy
Tel: +39-02-5666-0274
Email: italy@macrogen-europe.com